

MANIKANTH GOUD GURUJALA

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PROFILE SUMMARY

- **Machine Learning** engineer with hands-on experience building modular, robust **software systems** in **Python**: end-to-end ML pipelines and **LLM**-powered retrieval systems (**RAG**, **GraphRAG**) deployed to the cloud with evaluation and monitoring.
- Experienced across the **ML lifecycle** from **prototyping to production**, applying **software engineering best practices**: automated **testing**, **code reviews**, **CI/CD**, and **Git**.
- UBC Master of Data Science graduate (June 2026) with coursework in **time series forecasting** and **causal inference**, strong **SQL** foundations, and hands-on depth in **deep learning frameworks** (**PyTorch**) and **LLM** orchestration.

EDUCATION

Master of Data Science

Sept 2025 – Jun 2026

The University of British Columbia (UBC), Vancouver, BC

Relevant Courses: Machine Learning, Big Data Systems, Time Series Analysis, Causal Inference.

Bachelor of Mechatronic Engineering with Honors

Nov 2019 – Mar 2024

Asia Pacific University of Technology and Innovation (APU), Kuala Lumpur, Malaysia

WORK EXPERIENCE

Data Scientist - VRIFY (Capstone), Vancouver, BC

Apr 2026 – Jun 2026

- Owned and delivered a project **end to end**: from data pre-processing and exploration through **feature engineering** (reduced 360+ features via **RFE** and multicollinearity analysis), **offline experimentation** benchmarking five architectures, to packaging the final model for **production deployment**.
- Identified a hidden flaw in an initial evaluation setup producing misleadingly strong results, diagnosed the root cause (spatial autocorrelation), and implemented rigorous **validation** as a quality gate; documented design decisions and trade-offs for team knowledge sharing.
- Built scalable **data pipelines** on **Databricks** ingesting multi-source datasets with client **data anonymisation** protocols, supporting model training and **big data analytics** workflows.

AI/ML Intern, UltraTech Cement, Mumbai, India

Oct 2024 – Dec 2024

- Deployed **YOLOv8** inference models for **real-time production** monitoring across multiple industrial sites, achieving 88%+ detection accuracy and improving response times by 30%.
- Built anomaly detection models and **monitoring dashboards** with observability into model performance and costs, cutting costs by 15% and manual reporting time by 40%.

TECHNICAL PROJECTS

CodeIQ: Source Code Knowledge Graph, HCLTech × UBC AI/ML program

July 2026

- Built a code intelligence tool for **HCLTech** that parses a 100K-line React codebase into a queryable **knowledge graph** (3,820 entities, 25,000 edges) via AST extraction, letting developers and **LLM agents** query code structure and change impact in plain English.
- Layered a ChromaDB semantic **vector index** over the graph so answers cite real file paths and code snippets; packaged the full pipeline as a one-command, resumable **Docker** Compose workflow.

Hybrid RAG Movie Retrieval System (983K Amazon Reviews)

April 2026

- Built a hybrid **RAG** pipeline over 983K Amazon Movies & TV reviews (1.6M indexed chunks): **BM25** and **FAISS** rankings fused via Reciprocal Rank Fusion, answers generated by Llama 3.3 70B (Groq) with a web-search fallback; **deployed live on Streamlit Cloud**.

North American Crime Dashboard, UBC

March 2026

- Built and **deployed** an interactive **Shiny** for Python dashboard of North American crime trends (1975–2015) with a geocoded map and an AI tab for natural-language queries of the dataset via **Anthropic's Claude**.

ML Infrastructure & CI/CD: Open Source Python Package, UBC

Feb 2026

- Designed a modular **Python** package for data contracts (schema, missingness, drift) with Codecov-tracked tests, an automated **GitHub Actions** release pipeline to TestPyPI (Python 3.10–3.13), and hosted docs; enforced **code review** across the team.

TECHNICAL SKILLS

- **Software Engineering**: **Python**, modular system design, automated **testing** (unit/integration), **code reviews**, coding standards, **source control** (Git), **microservices** (**FastAPI**), technical **documentation**.
- **ML & Deep Learning**: **PyTorch**, **Scikit-learn**, **XGBoost**, **LightGBM**, **CatBoost**, **Optuna**, **CNNs**, **Transformers**, **Time Series Forecasting**, **Causal Inference**, **Feature Engineering**, **A/B Testing**, **SageMaker AI**.
- **DevOps & MLOps**: **CI/CD** (GitHub Actions), **Docker** containers, **Model Registry**, **MLflow**, **AWS**, **Claude Code**.
- **Data & Databases**: **SQL**, **MongoDB**, **Databricks**, **PySpark**, **DuckDB**, **Parquet**, **GraphDB**, **VectorDB**, **Big Data Ana-**

lytics.

- **Visualization:** Streamlit, Shiny (Python/R), Altair.
- **LLM Systems:** LLM APIs (Anthropic), **RAG** (BM25 + FAISS), **GraphRAG**, LangChain, LangGraph, Prompt Engineering.